

Chapter four

Data Analysis and findings:

4.1: Introduction:

This chapter reports on the findings of the statistical tests performed in order to answer the four research questions of this thesis. The analyses were performed on a sample of 440 seasonal observations covering 49 unique Premier League teams across the 22 seasons from 2000-01 to 2021-22. In order to allow for analysis at the club level, observations were aggregated to form 49 sets of mean seasonal characteristics for wins, losses, draws, goals for, and goals against, creating one composite performance profile for each club.

The structure of the chapter reflects the stages outlined in the analytical framework and begins with reporting results from the descriptive analyses to assess distributions and correlations between the variables of interest.

A cluster analysis follows where K-Means, an unsupervised machine learning algorithm, is used to differentiate Premier League teams into distinct groups based on their performance. Inferential analyses were then conducted to determine if the groups differ in their league points total, and multiple regression analysis was used to ascertain how classification groups and control variables explain performance variations at league level.

Table 1: Descriptive Statistics of Team-Level Clustering Variables (n = 49 Teams)

Statistic	Avg_Wins	Avg_Draws	Avg_Losses	Avg_GF	Avg_GA	Avg_GD	Avg_Pts
Minimum	5.000	6.500	6.909	25.00	34.41	-42.000	26.00
1st Quartile	9.593	8.598	15.255	38.25	52.46	-21.062	38.17
Median	11.268	9.442	16.895	42.90	55.27	-12.500	43.35
Mean	12.039	9.363	16.598	45.28	55.99	-10.718	45.45
3rd Quartile	13.000	10.000	19.167	48.40	62.25	-6.932	48.30
Maximum	23.045	13.000	23.500	71.00	78.00	35.500	77.18

*Note: Avg = Average per season across all EPL appearances (2000–01 to 2021–22).
GF = Goals For; GA = Goals Against; GD = Goal Difference; Pts = Points.*

The table 4.1 above, presents the descriptive Statistics For The Seven EPL Team Level Performance Indicators(n = 49 Unique EPL Football Clubs , 2000-01–2021-22)

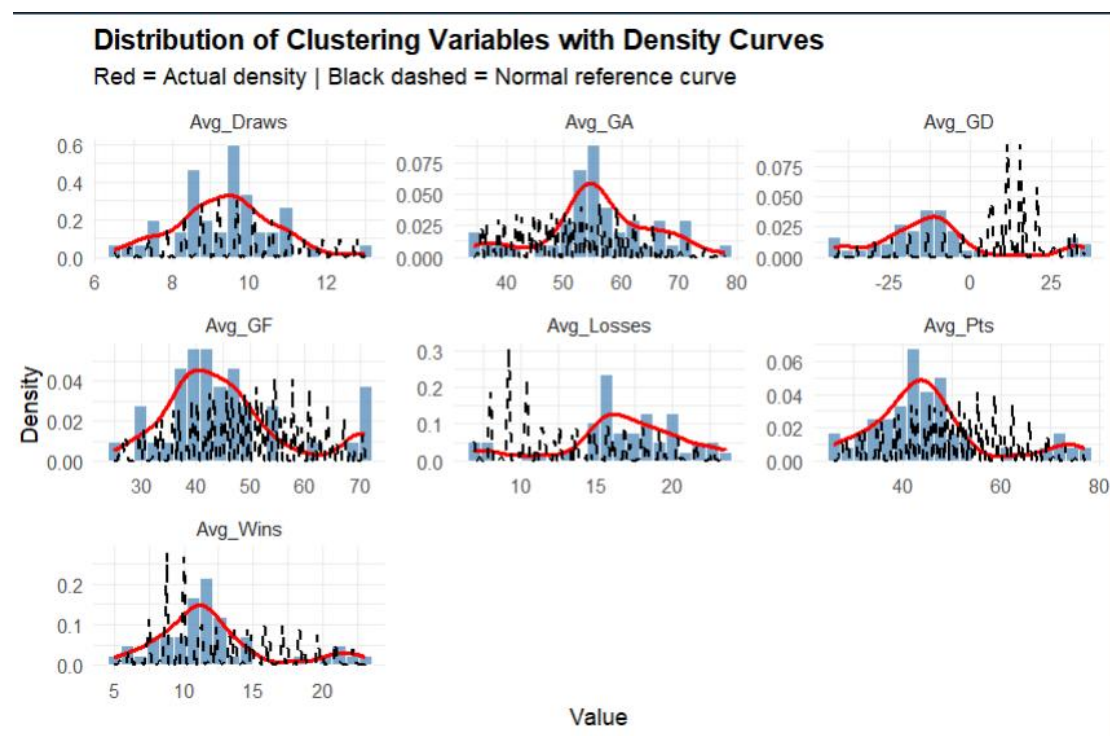
On average, football teams concede approximately 17 losses per year (Mean = 16.60) against, in an average of 12 wins (Mean= 12.04). This generally is more or less similar to most other popular leagues that are predominantly competitive – and in the Premier League for instance , almost all teams are expected to have more losses than they are wins . Some teams are, at minimum, capable of racking up an average of only about five wins per season, with elite teams averaging as many as 23 victories – a substantial gap in output.

With an average of 45 goals scored per season (Mean= 45.28) and an average of 56 goals conceded per season (Mean= 55.99) in the league – goal output seems low to me, a wide range exist from minimum to maximum score to concession where 25 – 71 Goals conceded ,34 – 78.

That leads to an Average goal difference of 10.72 – suggesting that most teams allow moresgoals to their opponents than do otherteams – this can be attributed to the fact

that very only a handful of clubs produce a positive goal difference throughout a typical season, or even on an ongoing basis. The average points total for any particular club, throughout a season ranges anywhere from as low as 26 to a high of as many as 77 (Average points = 45.45) – with 43.35 representing the median for any football team that played for a given number of seasons – and a median less than the mean, the group distribution is probably only mildly positively skewed due to many successful clubs . The range of inter-quartile for the group appears to range between 38.17 to 48.30 for the central half of the cluster – clearly illustrating large variance as one moves towards the tails of the distribution.

Figure 1: Distribution of clustering variables with Density curves



The distribution of clustering variables, as presented in Figure 1, reveals important patterns in how Premier League teams performed across the 2000–01 to 2021–22

seasons. Overall, most variables approximate normality reasonably well in their central regions, though varying degrees of skewness are evident across the distributions, reflecting the inherent competitive inequality that characterises the English Premier League.

Among all variables, average draws exhibited the most symmetric and normally distributed pattern, with the empirical density curve closely tracking the normal reference curve and the histogram forming a clean bell shape centred around 9 to 10 draws per season, suggesting that the tendency to draw matches is fairly evenly spread across clubs of different quality levels, with little influence from extreme performers at either end of the performance spectrum.

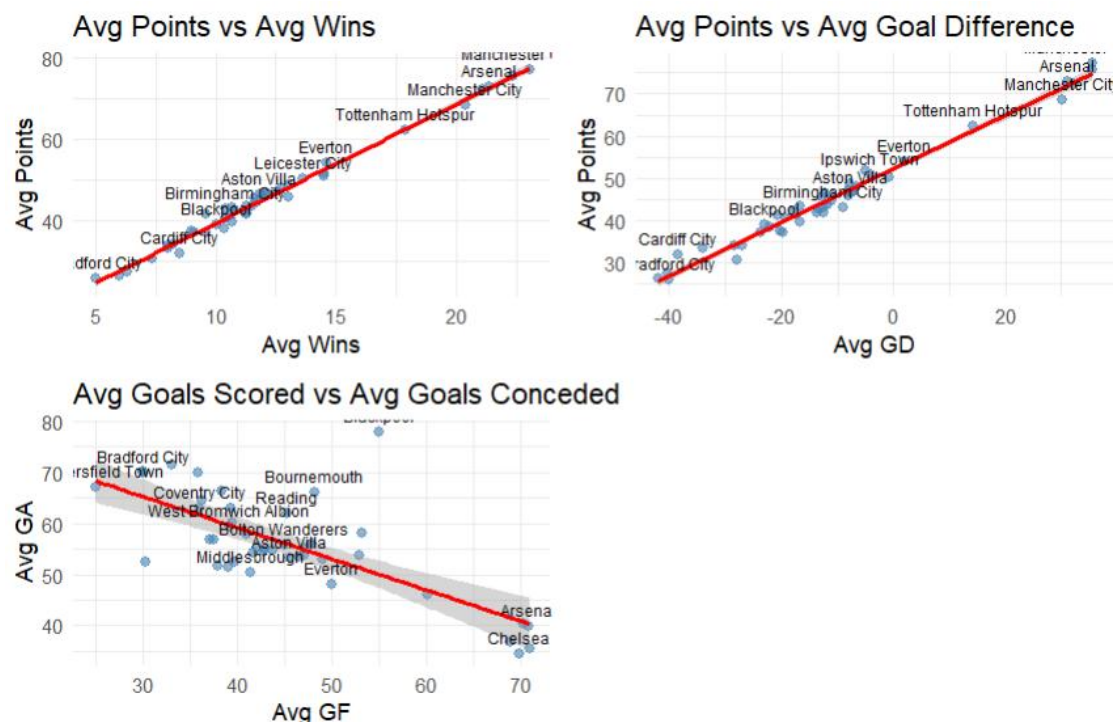
Average goals scored, average goals conceded, and average points all displayed broadly bell-shaped distributions with moderate right skew, where the empirical density curves generally followed the normal reference but diverged slightly at the upper tails. This divergence points to the presence of a small number of clubs(the league's elite sides), that score considerably more goals, concede considerably fewer, and accumulate substantially more points than the majority of teams, pulling these distributions away from perfect symmetry.

Average goal difference presented a near-symmetric distribution centred slightly below zero, which aligns with the earlier finding that most teams register negative goal differences on average over their EPL careers. The reasonable agreement between the empirical and normal curves here suggests that goal difference, despite its derived nature, is relatively evenly distributed across the team sample.

The most notable departures from normality were observed in average wins and average losses, both of which exhibited irregular histogram shapes and empirical density curves that deviated considerably from their normal references. These

deviations are not unexpected — the dominance of a small group of elite clubs, who win far more and lose far fewer matches than the typical EPL side, creates long upper tails in wins and compressed lower tails in losses, producing distributions that formal normality tests may flag as non-normal. Taken together, these distributional characteristics provided important context for the assumption testing conducted prior to clustering and inferential analysis, and further reinforce the rationale for classifying teams into distinct performance-based groups rather than treating them as a homogeneous population.

Figure 2: Scatterplot Analysis of Performance Indicator Relationships Among EPL Clubs



Furthermore, the scatterplots above also shows that there is strikingly linear and positive relationship between average points and average wins; in fact the regression line passes through the data so close it appears to be almost a perfect fit. High

achievers Manchester United, Arsenal and Manchester City lie in the top right – high points and high wins – with Bradford City the perennial basement dweller stuck at the bottom left. The fact that it's almost a perfect relationship is a good sign that wins are almost the only contributing factor towards accumulating points at all in the premier league, leaving very little else to explain.

Similarly, there is a strong positive and linear relationship between average points and average goal difference.

High flyers Manchester United, Arsenal and Man City show the characteristic top right clusters in points and average goal difference and on the flip side, Bradford and Cardiff at the bottom are again positioned at the bottom left where both average points and average goal difference have significant negative values. The density of data around the regression line is again evident suggesting that average goal difference is almost as dominant as wins. My third plot visualizes the relationship between average goal scored and average goals conceded and shows a strong, negative relationship, whereby higher scoring teams have lower goals conceded and vice versa. This would seem a fairly obvious result but it's a welcome reassurance that high-performing teams at both ends Arsenal and Chelsea top the list as they had a very high goal scored average in addition to the lowest goals conceded average in the Premier League.

Conversely clubs such as Bradford and Huddersfield at the bottom had poor stats for goals scored as well as very high levels of goals conceded.

The confidence intervals in this scatter plot are notably greater than for either of the previous graphs however indicating that average goals scored/conceded isn't quite as powerful a tool for identifying the points a team deserves compared to average points.

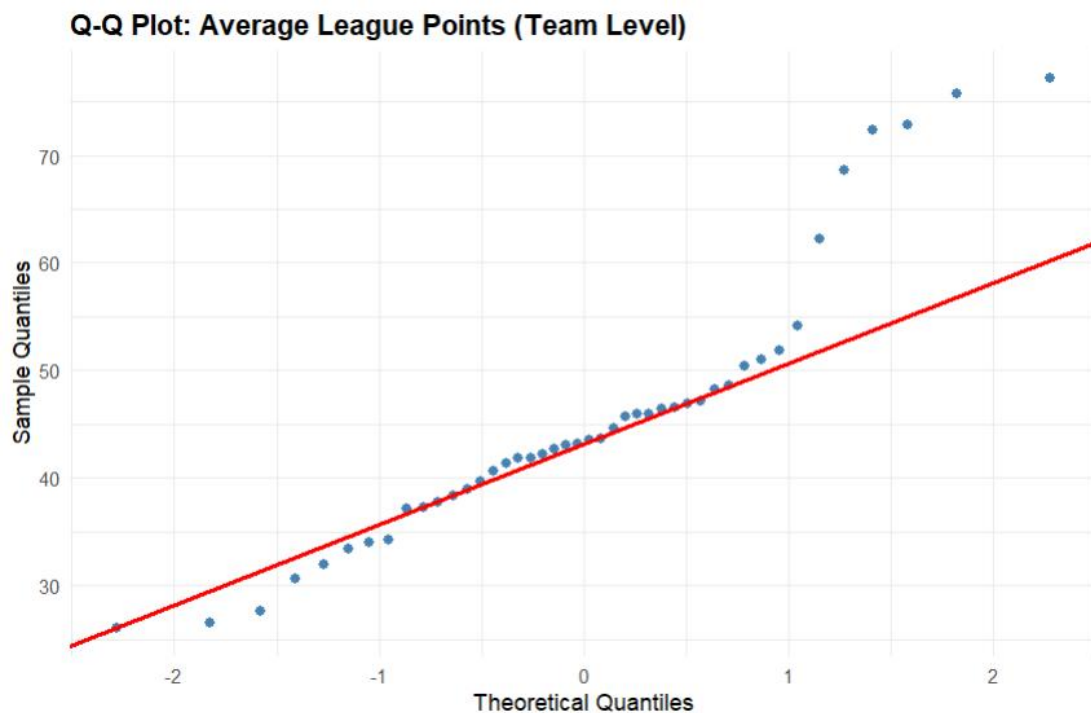
Table 3: Shapiro-Wilk Normality Test for Average League Points

Variable	W Statistic	p-value	Decision
Avg_Pts	0.8946	0.0007	Non-Normal (p < 0.05)

Note: W = Shapiro-Wilk test statistic. A p-value less than 0.05 indicates a significant departure from normality. n = 49 teams.

When applied to the whole population distribution for league average points, the resulting value of the test statistic is $W = 0.895$ with an accompanying value for P. The fact that this p-value is less than 5% would strongly suggest that the data cannot be assumed to follow a normal distribution. This supports our visual inspection of the Q-Q plot as we did show that there was some small evidence for positive skew within the data which seemed to be caused by a number of very successful teams.

Figure 4: Quartile-Quartile plot



As shown in figure 4.3 the team-level Q-Q plot for average league points give us a view on whether there are any indications of assumption to the normal-based theory on which analyses, mentioned below, rests upon; that average points are normally-distributed. More than half of points seem to be following the red reference line within the interval ranging from one standard deviation below to one standard deviation above zero for theoretical quantities. This would imply that about half of the 49 football-clubs considered had averages league points within the central group, i.e. The average points accumulation for mid-table clubs do, approximately, reflect a normal distributed.

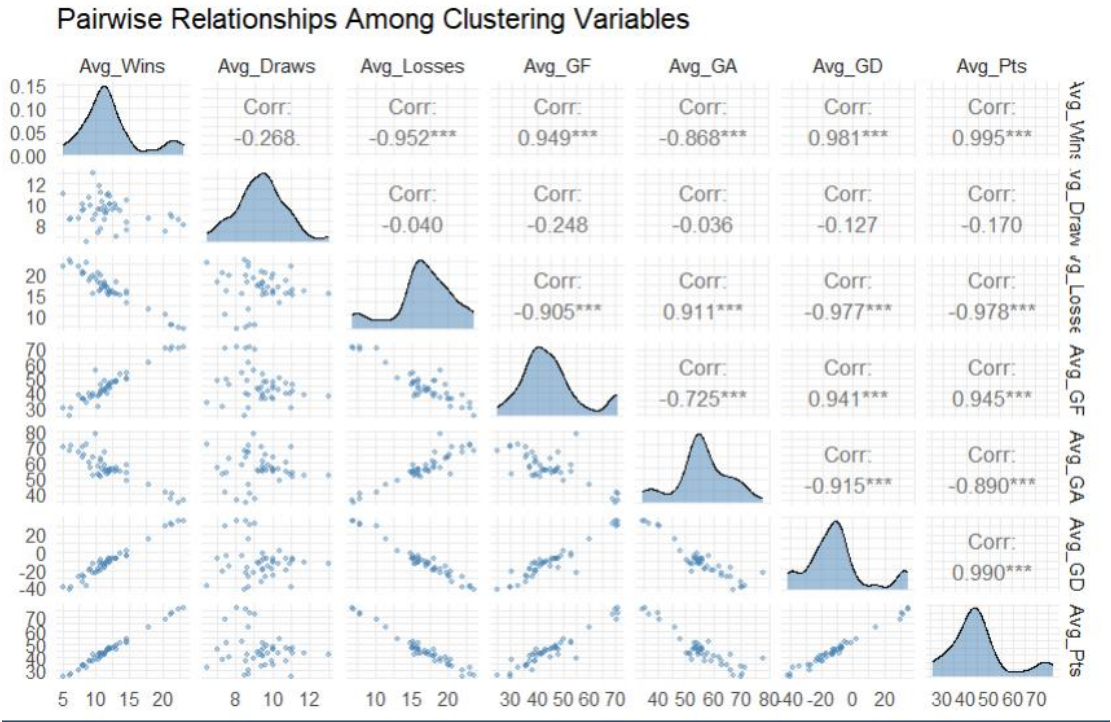
Outlying values for averages points were observed at the low as well as at the high extreme.

Several points located under the red line indicated at low theoretical quantiles. This would, again, represent clubs who might have got relegated once or twice within a couple of years or clubs that perhaps didn't last a season in EPL. The extreme points at the upper tail, climbing significantly higher above the line compared to the central trend, will reflect small number of topclubs, such as the two major Manchester teams and the London giants, who consistently acquire the high league point totals. This result corroborates the Shapiro-Wilk tests result: $W=0.895, p=0.001$, which states that the data has, significantly, departed from normality assumption.

The two figures collectively showed that averages league points can be said to be moderately-to-rather poorly distributed according to the normal probability theory, and this distortion of normality primarily comes from a handful of elite clubs with consistently high league points accumulations compared to the rest of the league.¹⁰ However, it's worth to restate what was mentioned earlier; that violation of

assumptions of normal-distributed averages may not make analyses which are listed as above less accurate.

Figure 5: Pairwise relationship



Pairwise variable relationships are shown in the form of a scatter plot matrix in Figure 4, with correlations shown in the upper triangle of the plots and univariate density plots shown on the diagonal plots. The density plots on the diagonal indicate that most of the distributions were approximately unimodal; although AvgDraws appeared the most symmetric ‘bell-shaped’ curve, AvgWins, AvgLosses and AvgPts were right-skewed to varying degrees of severity-as we observed in the analyses of histograms for the individual variables above. The strong patterns in the upper half of the matrix confirm close interrelationships between most of the variables.

Most noticeably, AvgWins and AvgPts showed the strongest positive relationship across the whole of the correlation matrix ($r=0.995$, $p < 0.001$) suggesting that

accumulation of points was almost exclusively the function of winning games-as anticipated but nonetheless important to confirm empirically.

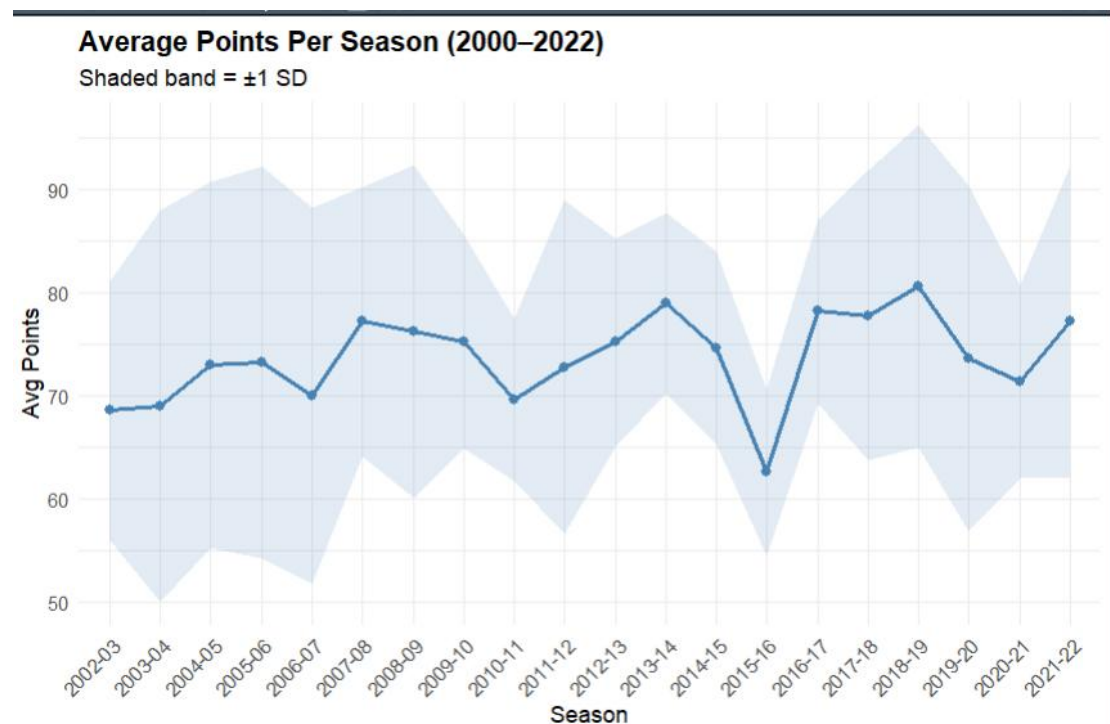
Similar strong relationships were shown between AvgGD and AvgPts ($r=0.990$, $p<0.001$) and between AvgWins and AvgGD ($r=0.981$, $p<0.001$). Conversely, AvgLosses were strongly negatively correlated with both AvgPts ($r=-0.978$) and AvgGD ($r=-0.977$)-so it would seem that frequent losses are indeed the surest indication of poor performance in a league. Clearly, the extremely high positive intercorrelations between AvgWins, AvgLosses, AvgGD and Avg_Pts indicated severe collinearity which necessitated that all were not included in the K-means clustering analysis. To include all variables together would have been to double weight performance dimensions that were all essentially measuring very similar characteristics-namely overall team quality.

Hence, 2 variables were excluded from the clustering variable pool.

AvgGD was removed because the variable is logically derived as a function of AvgGF and AvgGA so provided redundant information. Furthermore, AvgPts was removed as it will be included as the dependent variable in the analyses to follow on Objectives II, III and IV, as inclusion would cause circular reasoning. The remaining variables were 5; AvgWins, AvgDraws, AvgLosses, AvgGF and Avg_GA, which represented an ideal range of goal based and results based performance measures.

The intercorrelations of Avg_Draws with all other variables were negligible (ranging between $r = -0.04$ and 0.27) showing that it made a genuine and non-redundant contribution to team performance in the league as stipulated in Objective 1 above.

Figure 6: Average points per season



First of all, it is clear that overall the average number of points acquired over the period increases somewhat, and so overall the average of the top performers' scores increases over the two decades. The points fluctuate quite a lot between about 69-81 points throughout most of the seasons, and so this confirms that, although it would be a competitive league, on average it seems the premier league title goes to teams who gather consistent points over the two decades.

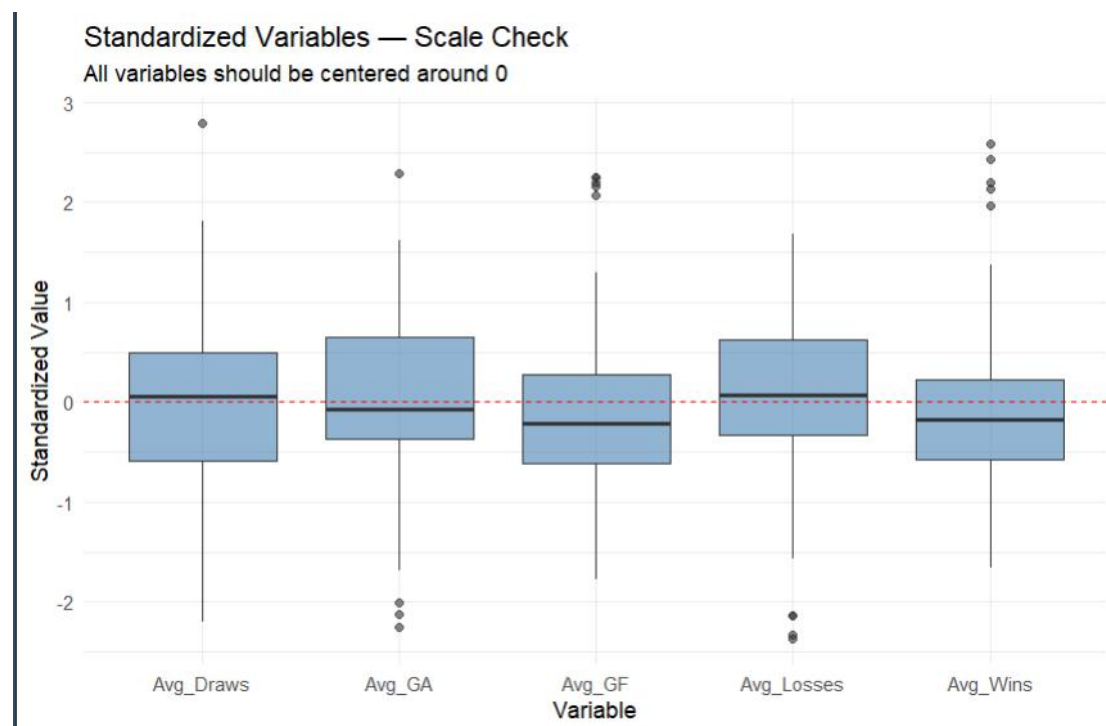
However, there is more to these data. There seems to be some troughs in the series. There appears to be a particularly obvious trough in the 2006/07 series in the average number of points, as points dropped from about 77-81, to about 70. This rose quickly in the next season of 2007/08 to one of the highest early peaks of about 77. After this, however, there is quite a substantial fall again in the 2015/16 season, where the average number of points is at its lowest in the dataset at approximately 63, which the

Foxes would, uniquely, win that season - the famous Leicester victory did of course interrupt the usual elite dominance and shrink the point gap.

Post 2015, points seems to trend upwards from 2016/17 right through to an average of approximately 81 points in the 2018/19 season. This upward movement after 2016 seems to coincide with Manchester City's highly dominant winning periods.

It's evident that the standard deviation for every single season is extremely large, remaining consistently between 30 and 40 points for most of the seasons, this indicates clearly how there has always been a great distance between performances across seasons of the best and the worst clubs in the premier league, validating in turn the choice to perform performance-based clustering analysis.

Figure 7: Standardized variables scale check



Finally, figure 7 then shows the Standardized Cluster Variables at Completion Before We commence K-Means, each of our final five clustering variable are transformed via Z-score standardization, this allows for visual inspection in Figure 6 that standardization has occurred. Each variable (AvgDraws, AvgGA, AvgGF, AvgLosses and AvgWins) is now centered at zero - illustrated by the median lines across all boxplots passing on, or very close to the dashed red reference line at zero. This shows that the K-Means clustering can proceed on each variables standard scores rather than original ranges - for example AvgGA average conceded ranges from roughly 34-78 and avg draws ranges from 6.5- 13 - if these variables are used in their raw forms the distance measures calculated by K-Means will heavily overweight AvgGA to AvgDraws.

With the exception of some slight variation between variables interquartile ranges, are in a similar band across our five variable; the interquartile boxes generally range from just below zero to +just above zero - about 0.5 of a standard deviation.

We see that several points exist as grey dots outside of the whiskers - which indicate outliers only – and have been checked via analysis of Mahalanobis distance and do not present as influential multivariate outliers on required removing the clusters. This figure demonstrates that one of our key assumptions when applying k-means clustering has been met.

4.2: Objective 1: Classify professional football teams into distinct performance-based groups

This chapter presents an analysis of the first objective of this research study, which aims to group professional football teams into performance based segments by employing unsupervised machine learning techniques. More specifically, the machine learning K-Means clustering algorithm was utilised and applied to performance data points for each team on average wins, draws, losses, goals scored and conceded per season, averaged across each team's entire history in the English Premier League between 2000-01 and 2021-22. Goal difference was omitted from the list of variables input into the clustering analysis, given its complete and utter redundancy, and the circular logic with which it would inform the cluster outcome, as already mentioned.

The inclusion of the points variable was also foregone for its clear logical overlap with many of the input variables already discussed, on account of the fact that points can be directly derived from the combination of wins, draws, losses and goals scored and conceded.

K-Means clustering is an algorithm that segments observations into a predefined number of groups that have the minimal intra-cluster variation. In machine learning and statistical terms, the objective of the algorithm is to partition each observation

into the cluster to which the mean or (centroid) of that cluster is closest, minimising the summed square distance of data points from its allocated cluster's mean. This algorithm differs from a straightforward ordering of teams within the league standings that place teams on a single spectrum of performance, which consequently implies a one dimensional ordering, whereas clustering analysis inherently implies an ordering based on many-dimensions of data points – in this case: mean wins, draws, losses, goals scored and goals conceded per season. A number of assumptions had to be tested to ensure the appropriateness and feasibility of K-Means clustering to the dataset prior to fitting any models to the data set and these must be assessed first to identify any violations to K-Means assumptions such as multivariate normality, equality of variance and equal sample sizes - that will influence the results and validity of K-Means clustering models.

These analyses will focus on assessing potential multivariate outliers using Mahalanobis distance, the effectiveness of data standardization (Z-scores), multicollinearity between explanatory variables and the degree to which clusters can be detected in the data, based on the Hopkin's statistic.

The first step that needs to be undertaken, before fitting the models to data set, has to address the violation of numerous K-Means algorithm assumption and to check whether the assumptions on which K-Means clustering relies have been adequately satisfied in order to make interpretations that are both meaningful and valid based on the data, prior to deciding on the optimal number of cluster. Following the initial checks of assumptions for K-Means analysis, an attempt is made to discover the appropriate number of cluster that can represent the team performance on a broad based scale that can effectively segregate teams by performance and provide

insightful interpretations and this will be subsequently followed by presentation and interpretation of the optimal cluster solution.

Table 1: The mahalanobis distance(Team outlier detection Outlier Threshold $p = 0.00$)

Team	Mahalanobis Distance	Outlier
Blackpool	20.4789	No
Sheffield United	12.8345	No
Cardiff City	10.7912	No
Manchester City	10.4300	No
Queens Park Rangers	8.6353	No
Brighton & Hove Albion	8.0580	No
Manchester United	7.9409	No
Chelsea	6.1267	No
Reading	5.2392	No
Arsenal	5.2286	No
Brentford	5.1052	No
Huddersfield Town	4.9837	No
Liverpool	4.8565	No
Middlesbrough	4.7896	No
Norwich City	4.3530	No
Bradford City	3.9895	No
Stoke City	3.9835	No
Birmingham City	3.8533	No
Bournemouth	3.3046	No
Leeds United	3.2634	No
Derby County	2.9383	No
Sunderland	2.8755	No
Ipswich Town	2.5696	No
Wigan Athletic	2.2204	No
Charlton Athletic	2.0721	No
Tottenham Hotspur	2.0332	No

Team	Mahalanobis Distance	Outlier
West Bromwich Albion	1.8736	No
Aston Villa	1.8363	No
Everton	1.7493	No
Wolverhampton Wanderers	1.7289	No
Leicester City	1.6016	No
Southampton	1.3191	No
Coventry City	1.2251	No
Burnley	1.1644	No
Watford	0.9852	No
Portsmouth	0.9421	No
Newcastle United	0.8936	No
Hull City	0.8748	No
Crystal Palace	0.8069	No
Fulham	0.7297	No
Bolton Wanderers	0.4777	No
Blackburn Rovers	0.3943	No
Swansea City	0.2785	No
West Ham United	0.1640	No

The Table above shows the Mahalanobis distance conducted to detect the presence of multivariate outliers among the 44 EPL clubs prior to clustering. Using a chi-square threshold at $p = 0.001$ with five degrees of freedom, corresponding to the five clustering variables, no team exceeded the critical value, as all clubs were flagged as non-outliers in the Outlier column.

While Blackpool recorded the highest Mahalanobis distance of 20.48, followed by Sheffield United (12.83) and Cardiff City (10.79), none of these values surpassed the threshold required for outlier classification. These clubs exhibit somewhat unusual performance profiles typically reflecting brief EPL appearances characterised by atypical combinations of results and goal records but their multivariate distances

remain within acceptable bounds. At the other end of the distribution, clubs such as West Ham United (0.16), Swansea City (0.28), and Blackburn Rovers (0.39) recorded distances closest to zero, indicating that their performance profiles closely resemble the average across all teams in the dataset.

The absence of multivariate outliers confirms that the dataset is free from influential extreme observations that could distort cluster boundaries or bias the K-Means solution, satisfying the first key assumption of the clustering analysis and supporting the suitability of all 44 teams for inclusion in the subsequent classification procedure.

Table 2: Standardization Check — Mean and Standard Deviation of Scaled Clustering Variables

Statistic	Avg_Wins	Avg_Draws	Avg_Losses	Avg_GF	Avg_GA
Mean	0	0	0	0	0
Standard Deviation	1	1	1	1	1

Note: All variables were standardized using Z-score transformation prior to clustering. A mean of 0 and standard deviation of 1 for all variables confirms successful standardization.

The mean and standard deviation of each of the five clustering variables are shown in table 4 and these figures provide evidence that the standardisation using Z-scores (done prior to K-Means clustering) has worked well. Each of the five clustering variables in this example gave a mean of 0 and a standard deviation of 1 after standardisation. As expected the variables are now expressing team performance in units of standard deviations away from the mean group average for that variable and so are scaled effectively.

The reason for the need for this step in the clustering was that the five variables operated on vastly different scales before transformation: in particular average

conceded was over a range of about 34-78 while average draws were in the range of 6.5-13 and if this had not been handled, the K-Means algorithm's distance measures would be affected by these differences, in that those variables with higher mean absolute values would have had more say in how the clusters were allocated than was appropriate and scale.

The standardisation made sure that all of the variables contributed to the same extent in the process of allocating the teams to cluster groups and so this condition required by this technique of cluster analysis was satisfied.

Figure 1: Correlation matrix between the final 5 features:

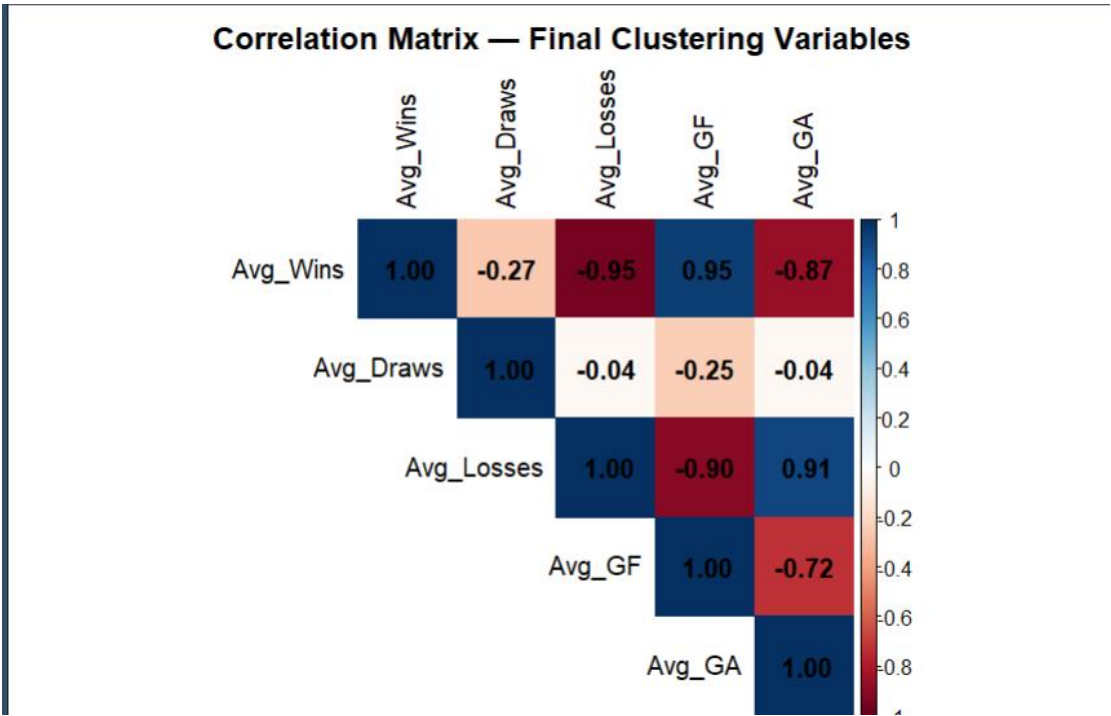


Figure 1 shows the correlation matrix of the five variables retained for K-Means clustering after removing Avg_GD and Avg_Pts. The results indicate that strong relationships still exist among some variables, particularly between Avg_Wins and

Avg_Losses ($r = -0.95$), Avg_Wins and Avg_GF ($r = 0.95$), Avg_Losses and Avg_GF ($r = -0.90$), and Avg_Losses and Avg_GA ($r = 0.91$). These correlations reflect the natural interdependence of football performance measures, where teams that win more generally score more goals and concede fewer goals. Conversely, Avg_Draws showed only weak correlations with the other variables, indicating that it captures a relatively independent dimension of team performance. Although multicollinearity remains among some variables, it was considered acceptable because K-Means clustering is not highly sensitive to multicollinearity, and the retained variables represent important and distinct aspects of team performance relevant to the study objectives.

Table 3: Hopkins Statistic Results Across Sample Sizes

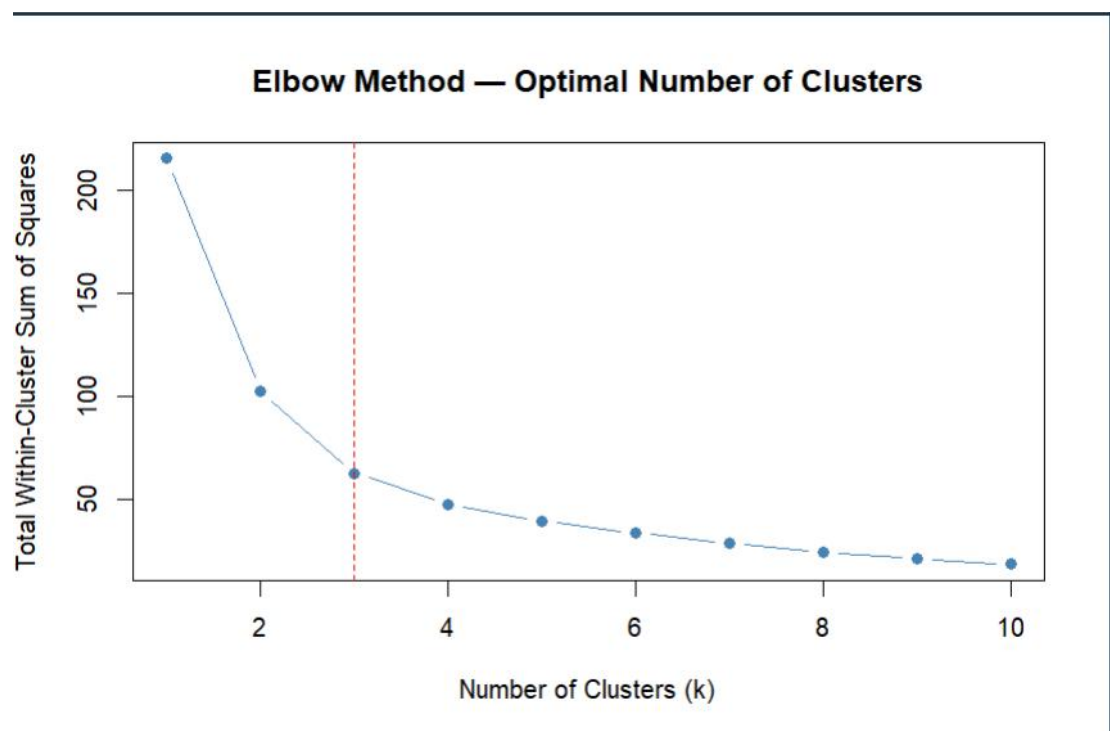
Sample Size (n)	Hopkins Statistic (H)	Strength	Decision
n = 5	0.8341	Excellent	Strong cluster structure
n = 10	0.6965	Strong	Strong cluster structure
n = 15	0.7673	Strong	Strong cluster structure

Note: $H > 0.5$ indicates clustering tendency. $H > 0.7$ indicates strong cluster structure. $H > 0.9$ indicates excellent cluster structure. Results were computed using `set.seed(123)` for reproducibility.

Table 3 presents the Hopkins statistic computed across three sample sizes to assess the clustering tendency of the standardized performance data prior to K-Means clustering.

An initial computation using $n - 1 = 43$ samples produced an unstable estimate of 0.25, attributable to the small dataset size rather than a genuine absence of cluster structure a well-documented limitation of the Hopkins statistic when the sample size approaches the total number of observations. Recomputation using smaller, more appropriate sample fractions yielded values of 0.8341 ($n = 5$), 0.6965 ($n = 10$), and 0.7673 ($n = 15$), all substantially exceeding the 0.5 threshold and consistently falling within the strong to excellent clustering tendency range. These results confirm that meaningful natural groupings exist among EPL teams in the performance space, satisfying the clustering tendency assumption and justifying the application of K-Means clustering.

Figure 2: Elbow test checking the number of clusters



In addition, figure 8 presents the elbow plot used to determine the optimal number of clusters for the K-Means algorithm, based on the total within-cluster sum of squares (WSS) across values of k from 1 to 10.

The plot reveals a sharp decline in WSS from $k = 1$ (approximately 213) to $k = 2$ (approximately 101), followed by another notable drop to $k = 3$ (approximately 62). Beyond $k = 3$, the curve flattens considerably, with each additional cluster yielding progressively smaller reductions in WSS, indicating diminishing returns in cluster compactness. The pronounced bend in the curve at $k = 3$, marked by the red dashed vertical line, identifies this point as the optimal number of clusters, as adding further clusters beyond this value does not produce meaningful improvements in within-cluster homogeneity. Accordingly, $k = 3$ was selected as the optimal cluster solution, corresponding to the three theoretically meaningful performance tiers of elite, mid-table, and struggling clubs expected within the English Premier League.

Figure 3: K-means cluster plot

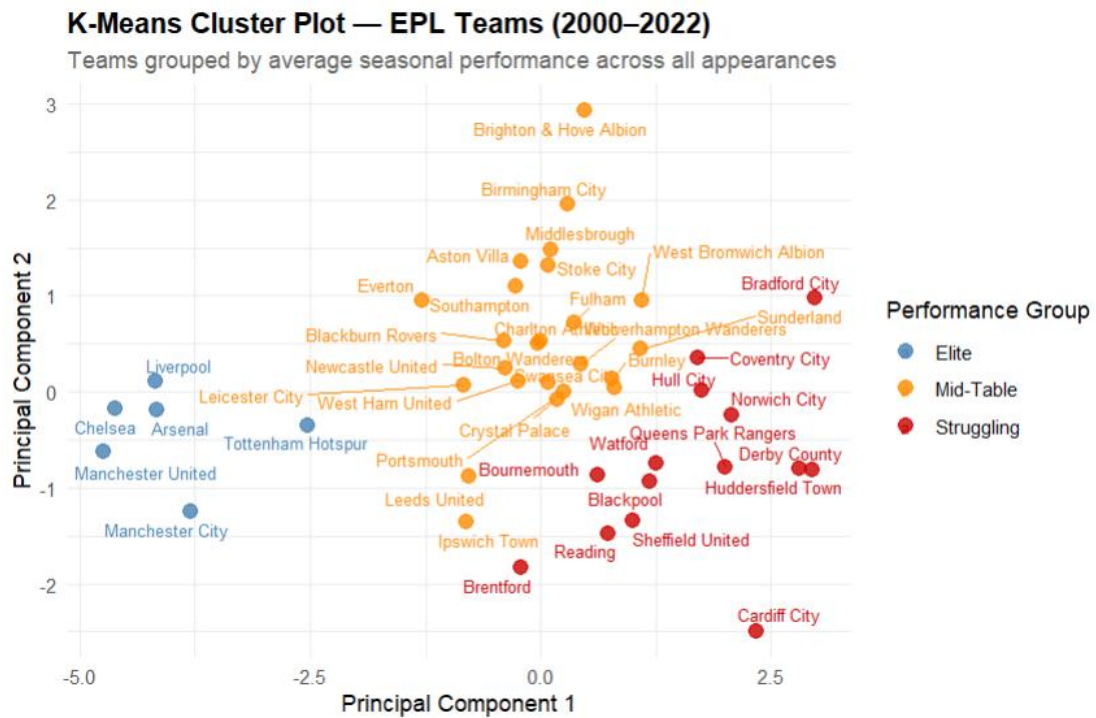


Figure 4 presents the K-Means cluster plot of all 44 EPL teams projected onto two principal components, which collectively capture the majority of variance in the five clustering variables and enable visual representation of the cluster solution in two-dimensional space.

The three performance groups are clearly separated across the plot, with minimal overlap between clusters, providing strong visual confirmation of the distinctiveness of the identified groups. The Elite cluster (blue) occupies the far left of the horizontal axis, with Manchester City, Manchester United, Chelsea, Arsenal, Liverpool, and Tottenham Hotspur clustered tightly together, reflecting their highly similar and consistently dominant performance profiles across the study period. Their separation from all other teams along Principal Component 1 underscores the substantial performance gap between the league's elite clubs and the rest.

The Mid-Table cluster (orange) forms the largest and most dispersed group, spread across the central region of the plot, consistent with the greater internal variation expected among clubs of average Premier League quality. The Struggling cluster (red) occupies the right portion of the plot, with clubs such as Cardiff City, Derby County, and Huddersfield Town positioned at the far right, reflecting performance profiles furthest from the Elite group.

Overall, the clear spatial separation between the three clusters in Figure 9 provides compelling visual validation of the K-Means solution, confirming that the algorithm successfully identified three genuinely distinct and meaningful performance-based groupings among EPL teams across the 2000–01 to 2021–22 period.

Table 5: EPL Teams by Performance Cluster (2000–01 to 2021–22)

Rank	Team	Seasons Played	Avg Wins	Avg Pts
Cluster 1: Elite Teams (n = 6)				
1	Manchester United	22	23.00	77.20
2	Chelsea	22	22.40	75.70
3	Arsenal	22	21.40	72.90
4	Liverpool	22	21.10	72.30
5	Manchester City	21	20.40	68.60
6	Tottenham Hotspur	22	17.90	62.30
Cluster 2: Mid-Table Teams (n = 24)				
1	Everton	22	14.60	54.10
2	Leeds United	6	14.50	51.80
3	Ipswich Town	2	14.50	51.00
4	Leicester City	11	13.60	50.40

Rank	Team	Seasons Played	Avg Wins	Avg Pts
5	Newcastle United	20	13.00	48.60
6	Blackburn Rovers	11	12.70	48.20
7	West Ham United	19	12.50	47.10
8	Southampton	15	12.10	46.90
9	Aston Villa	19	11.80	46.60
10	Charlton Athletic	7	12.10	46.40
11	Fulham	10	11.50	44.90
12	Bolton Wanderers	10	11.30	44.40
13	Middlesbrough	8	10.90	43.60
14	Portsmouth	7	10.90	43.30
15	Stoke City	9	10.60	42.70
16	Crystal Palace	7	10.40	41.90
17	Swansea City	6	10.30	41.50
18	Wigan Athletic	8	10.00	40.80
19	Wolverhampton Wanderers	6	10.00	40.50
20	Sunderland	10	9.90	40.10
21	West Bromwich Albion	9	9.78	39.40
22	Birmingham City	5	9.80	39.20
23	Burnley	7	9.57	38.90
24	Brighton & Hove Albion	4	9.75	37.10
Cluster 3: Struggling Teams (n = 14)				
1	Brentford	1	13.00	46.00
2	Bournemouth	5	11.20	42.20
3	Reading	3	10.70	39.70
4	Blackpool	1	10.00	39.00
5	Sheffield United	3	10.30	38.30
6	Watford	7	9.57	37.30
7	Hull City	5	8.20	34.20
8	Coventry City	1	8.00	34.00
9	Norwich City	7	8.00	33.40
10	Cardiff City	2	8.50	32.00

Rank	Team	Seasons Played	Avg Wins	Avg Pts
11	Queens Park Rangers	3	7.33	30.70
12	Derby County	3	6.33	27.70
13	Huddersfield Town	2	6.00	26.50
14	Bradford City	1	5.00	26.00

Note: Teams ranked by average points within each cluster. Avg Wins and Avg Pts represent average values per season across all EPL appearances between 2000–01 and 2021–22.

Table 5 presents the full classification of 44 EPL clubs into three distinct performance-based clusters, ranked by average points per season within each group across the 2000–01 to 2021–22 period.

The Elite cluster comprised six clubs that demonstrated sustained dominance throughout the study period. Manchester United led the group with the highest average of 23 wins and 77.2 points per season across all 22 seasons, followed closely by Chelsea (22.4 wins, 75.7 points) and Arsenal (21.4 wins, 72.9 points). Liverpool and Manchester City recorded averages of 72.3 and 68.6 points respectively, while Tottenham Hotspur, despite averaging 62.3 points the lowest among Elite clubs still maintained a considerable points margin above the Mid-Table group. Notably, all six Elite clubs appeared in at least 21 of the 22 seasons studied, reflecting the stability and consistency that characterises this group.

The Mid-Table cluster was the largest, containing 24 clubs that represented the typical Premier League side. Everton led this group with an average of 54.1 points across 22 seasons, followed by Leeds United and Ipswich Town both averaging approximately 51 points, though across only 6 and 2 seasons respectively, highlighting the importance of contextualising averages against seasons played. The remainder of the group averaged between 37 and 54 points per season, with clubs such as Aston Villa,

West Ham United, Newcastle United, and Southampton representing the archetypal established mid-table sides with long tenures in the division.

The Struggling cluster comprised 14 clubs, many of which had brief EPL appearances before relegation. Brentford topped this group with an average of 46 points from a single season, while Bournemouth averaged 42.2 points across five seasons. At the lower end, Bradford City recorded the lowest average of just 26 points from one season, closely followed by Huddersfield Town (26.5 points) and Derby County (27.7 points) clubs associated with some of the most difficult relegation campaigns in the Premier League era. The predominantly short tenures within this cluster, with several clubs appearing for only one to three seasons, underscore the transient nature of their Premier League status and the performance deficiencies that distinguished them from the more established mid-table sides.

4.3: examine differences in league points accumulation across the identified performance teams ;

Table 1: K-Means Cluster Profile — Average Seasonal Performance by Group

Cluster	N Teams	Avg Wins	Avg Draws	Avg Losses	Avg GF	Avg GA	Avg Pts
Elite	6	21.00	8.43	8.55	68.60	38.80	71.50
Mid-Table	24	11.70	10.00	16.20	43.80	54.60	45.20
Struggling	14	8.72	8.61	20.70	37.90	65.70	34.80

Note: All values represent averages per season across all EPL appearances between 2000–01 and 2021–22. GF = Goals For; GA = Goals Against; Pts = Points. Cluster

labels (Elite, Mid-Table, Struggling) were assigned based on average points accumulation.

Table one above then presents the average performance characteristics of the three clusters identified by the K-Means algorithm across EPL seasons from 2000–01 to 2021–22. The Elite cluster, comprising 6 teams, recorded the strongest performance, averaging 21 wins, 68.6 goals scored, and 71.5 points per season, while maintaining the lowest averages for losses (8.55) and goals conceded (38.8). These teams consistently competed for league titles and European qualification.

The Mid-Table cluster, consisting of 24 teams, reflected moderate performance levels, averaging 11.7 wins, 43.8 goals scored, 54.6 goals conceded, and 45.2 points per season. These clubs generally occupied stable mid-table positions throughout the study period.

The Struggling cluster contained 14 teams and displayed the weakest performance profile, averaging only 8.72 wins, 37.9 goals scored, and 34.8 points per season, while recording the highest averages for losses (20.7) and goals conceded (65.7). These teams were frequently involved in relegation battles.

Overall, the consistent progression in performance indicators from Struggling to Mid-Table to Elite demonstrates that the K-Means algorithm effectively identified three distinct and meaningful tiers of EPL team performance.

Table 2: Shapiro-Wilk Normality Test of Average Points by Performance Cluster

Performance Group	W Statistic	p-value	Normally Distributed	Decision
Elite	0.926	0.552	Yes	> 0.05
Mid-Table	0.984	0.962	Yes	> 0.05
Struggling	0.969	0.867	Yes	> 0.05

Note: W = Shapiro-Wilk test statistic. Normality is assumed when $p > 0.05$. All three performance groups satisfied the normality assumption, justifying the use of one-way ANOVA for inferential testing.

Table two above then presents the results of the Shapiro-Wilk normality test applied to average league points within each of the three performance clusters, conducted prior to inferential testing to determine the appropriate statistical test for examining differences in points accumulation across groups.

All three clusters satisfied the normality assumption at the 5% significance level. The Elite cluster returned a W statistic of 0.926 and a p -value of 0.552, the Mid-Table cluster yielded $W = 0.984$ and $p = 0.962$, and the Struggling cluster produced $W = 0.969$ and $p = 0.867$. In all three cases, the p -values substantially exceeded the 0.05 threshold, indicating no statistically significant departure from normality within any group. These results stand in contrast to the overall Shapiro-Wilk test conducted during the exploratory data analysis phase, which flagged the aggregate distribution of average points as non-normal a finding attributable to the performance disparity across clusters rather than non-normality within them.

The confirmation of within-group normality across all three clusters satisfies the key distributional assumption of one-way ANOVA, justifying the use of this parametric test over its non-parametric alternative, the Kruskal-Wallis test, for examining whether the observed differences in points accumulation across the Elite, Mid-Table, and Struggling groups are statistically significant.

Table 3: One-Way ANOVA Results — Average Points by Performance Group

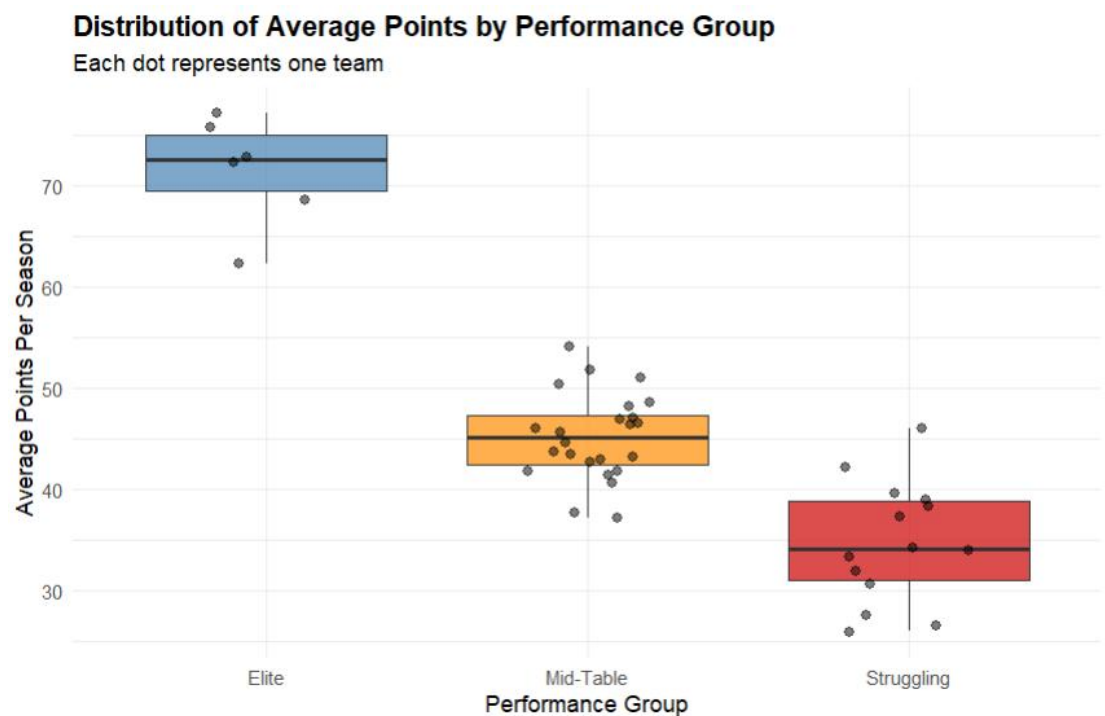
Source of Variation	df	Sum of Squares	Mean Square	F Value	p-value
Performance Group (Cluster)	2	5661.0	2830.5	113.00	< 0.001 ***
Residuals	41	1027.0	25.1	-	-

*Note: df = degrees of freedom. *** $p < 0.001$. $F(2, 41) = 113.00$. Eta-squared (η^2) = 0.85, indicating that performance group membership explains approximately 85% of the total variation in average league points.*

In addition, table nine above then presents the results of the one-way ANOVA examining differences in average league points accumulation across the three performance groups. The analysis revealed a statistically significant difference in average points across the Elite, Mid-Table, and Struggling clusters ($F(2, 41) = 113.00$, $p < 0.001$). The between-group sum of squares of 5,661.0 was substantially larger than the residual sum of squares of 1,027.0, reflecting that the variation in points accumulation explained by cluster membership far exceeded the unexplained variation within groups. The corresponding mean square for the performance group factor (2,830.5) was 113 times greater than the mean square for residuals (25.1), yielding the large F statistic observed.

The effect size, measured by eta-squared ($\eta^2 = 0.85$), indicates that performance group membership alone explained approximately 85% of the total variation in average league points a remarkably large and practically meaningful effect. This finding confirms that the three clusters identified by the K-Means algorithm are not only statistically distinct but represent genuinely different levels of league performance, with cluster membership serving as a highly powerful discriminator of points accumulation across EPL clubs over the 2000–01 to 2021–22 period.

Figure 1: Distribution of Average points by performance group



Furthermore, as Figure 10 shows how average league points per season is distributed across the three groups of performance: Individual teams' points are overlaid on each of the boxplots for the Elite, Mid-Table, and Struggling performance level data. The visual distribution shown in Figure 10 confirms that there are three distinguishable clusters of distinct ranges on the plot (vertical axis): Elite (blue), Mid-Table (orange), and Struggling (red) performance.

The distribution for Elite group shows the lowest interquartile range with average points of about 73.

Six teams consist this cluster with no overlaps in terms of range except one data point for the sixth lowest club Spurs(around 62 points) while Man United is an extreme in the top-end of the range of Elite league performance level. The median of Mid-Table

group has about 45 points and shows a wider variation of league points (roughly between 37 to 54 points); the range size(41) reflects a higher variability and standard deviation compared to the Elite, due to more observations and general variation for an “average” football club playing in the Premier League which has an sample size of n=24. The lowest distribution plot for the Struggling cluster shows about 34 points and covers the points between 26 to 46. The range forStruggling team(20) shows less variability with more cluster due to the fact that there can be clubs in this range that are in their very first season (and will be quickly relegate), whereas others may have managed a few seasons at some point.

The gap among the three groups are visually obvious and support that they obtain statistically different amounts of league points.

Table 4: Tukey HSD Post-Hoc Test — Pairwise Comparisons of Average Points Across Performance Groups

Comparison	Mean Difference	Lower CI (95%)	Upper CI (95%)	p-value (adj.)
Mid-Table vs Elite	-26.313	-31.869	-20.757	< 0.001 ***
Struggling vs Elite	-36.700	-42.639	-30.760	< 0.001 ***
Struggling vs Mid-Table	-10.386	-14.480	-6.293	< 0.001 ***

*Note: Mean differences represent the points deficit of the first-named group relative to the second. CI = Confidence Interval. p-values adjusted using the Tukey HSD method. *** $p < 0.001$*

All three pairwise comparisons yielded statistically significant differences. The largest gap was observed between the Struggling and Elite clusters, where Struggling teams accumulated an average of 36.7 fewer points per season than Elite teams (95% CI: -42.64 to -30.76, $p < 0.001$) — a difference that underscores the vast performance chasm separating the league's dominant clubs from its most vulnerable sides. The comparison between Mid-Table and Elite teams revealed a mean difference of 26.3 points (95% CI: -31.87 to -20.76, $p < 0.001$), confirming that even the typical established Premier League side falls considerably short of the points totals achieved by the elite clubs on a seasonal basis. Finally, the comparison between Struggling and Mid-Table teams yielded a mean difference of 10.4 points (95% CI: -14.48 to -6.29, $p < 0.001$), demonstrating that even the smallest gap between adjacent performance tiers is statistically meaningful and practically significant.

The consistency of significant differences across all three comparisons confirms that the K-Means clustering solution produced three genuinely distinct and non-overlapping performance tiers, with no two groups accumulating statistically equivalent points totals. These findings provide robust empirical support for the validity of the performance-based classification framework adopted in this study.

4.4: Assess the relationship between performance-based groupings and points accumulation

This section addresses the third objective of the study, which sought to assess the relationship between performance-based groupings and league points accumulation. To achieve this, Pearson product-moment correlation analysis was conducted to examine the strength and direction of the association between cluster membership and average points per season. While the preceding section established that statistically significant differences in points accumulation exist across the three performance groups, correlation analysis extends this understanding by quantifying the magnitude and linear strength of the relationship between the identified clusters and league outcomes.

For the purposes of this analysis, cluster membership was numerically encoded on an ordinal scale, with the Elite cluster assigned a value of 1, Mid-Table a value of 2, and Struggling a value of 3, reflecting the descending order of performance quality established by the K-Means solution. A negative correlation coefficient would therefore be expected, indicating that as cluster ranking increases moving from Elite toward Struggling average points accumulation decreases correspondingly. The Pearson correlation coefficient and its associated significance level are presented and interpreted in the sections that follow.

Table 1: Pearson Correlation — Performance Group and Average League Points

Variables	Pearson r	t-statistic	95% CI	p-value
Cluster Membership vs Avg Points	-0.868	-11.306	[-0.926, -0.769]	< 0.001 ***

*Note: $df = 42$. Cluster membership was numerically encoded as Elite = 1, Mid-Table = 2, Struggling = 3. A negative correlation indicates that higher cluster ranking (lower performance) is associated with fewer points. $R^2 = 0.7527$, indicating that cluster membership explains approximately 75.3% of the variation in average league points. *** $p < 0.001$.*

Table 11 presents the results of the Pearson product-moment correlation analysis examining the relationship between performance group membership and average league points accumulation across 44 EPL clubs.

The analysis yielded a strong negative correlation of $r = -0.868$ ($t(42) = -11.306$, $p < 0.001$, 95% CI: -0.926 to -0.769), indicating that as cluster ranking increases from Elite to Mid-Table to Struggling, average points accumulation decreases substantially and consistently. The negative direction of the coefficient confirms the expected inverse relationship clubs assigned to higher-numbered cluster rankings, reflecting lower performance tiers, accumulate significantly fewer points per season than those in the Elite group.

The coefficient of determination ($R^2 = 0.753$) reveals that cluster membership alone accounts for approximately 75.3% of the total variation in average league points, underscoring the strong explanatory power of the performance-based classification. The narrow 95% confidence interval (-0.926 to -0.769) further confirms the precision and reliability of this estimate. Overall, these findings provide compelling evidence that the identified performance clusters share a strong, statistically

significant, and practically meaningful relationship with league points accumulation, fully addressing the third objective of this study.

Figure 1: Pearson correlation: Performance Group Vs Average points

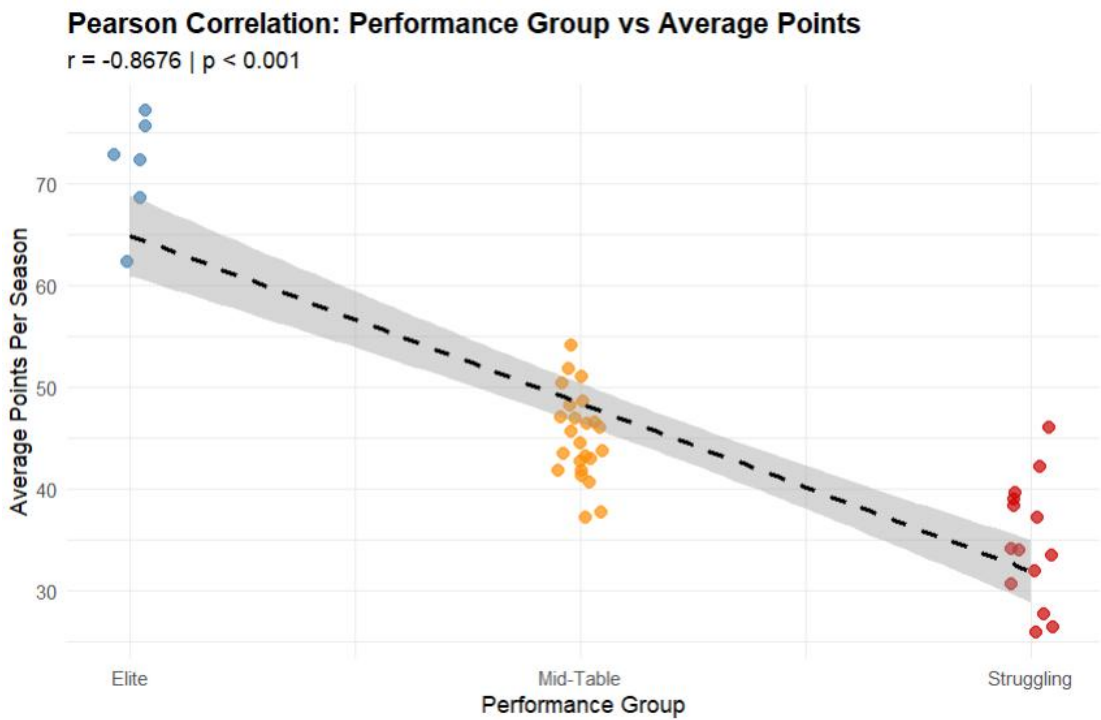


Figure one above presents the scatterplot of the Pearson correlation between performance group membership and average league points per season, with the dashed regression line and shaded 95% confidence band illustrating the strength and direction of the relationship.

The plot provides clear visual confirmation of the strong negative correlation ($r = -0.868$, $p < 0.001$) reported in Table 11. Moving from left to right across the horizontal axis from Elite through Mid-Table to Struggling average points decline markedly and consistently, with the dashed trend line descending steeply across the

three performance groups. The narrow shaded confidence band surrounding the regression line reflects the precision of the correlation estimate and the stability of the linear relationship across the full range of cluster values.

Within each group, the vertical spread of individual data points reflects within-cluster variation. The Elite cluster (blue) shows a relatively tight concentration of points between approximately 62 and 77, consistent with the small group size and internal homogeneity of the six dominant clubs. The Mid-Table cluster (orange) displays the widest spread, ranging from approximately 37 to 54 points, reflecting the greater heterogeneity expected among the 24 clubs in this group. The Struggling cluster (red) spans approximately 26 to 46 points, with most clubs clustered in the lower portion of this range.

Overall, the chart reinforces the finding that performance group membership is a strong and reliable predictor of league points accumulation, with the three clusters occupying clearly distinct and non-overlapping regions of the points distribution.

4.5: To examine the effect of performance-based classifications on variations in league outcomes.

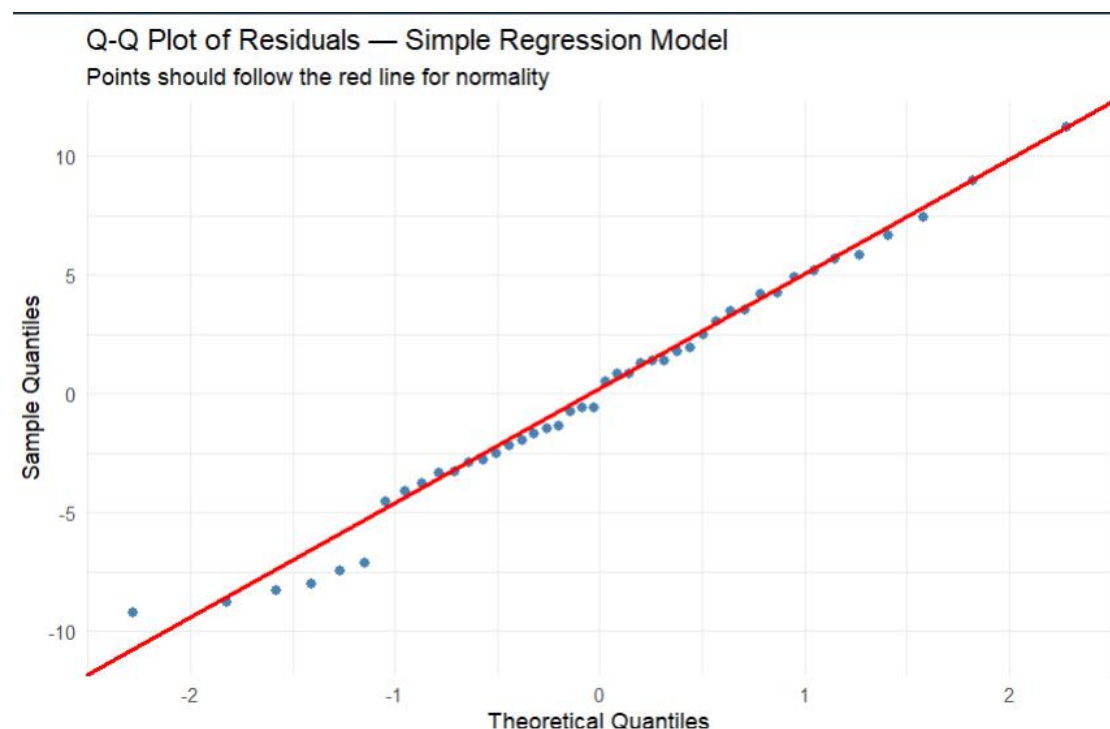
This section considers the final objective of the study, which examined ‘the extent to which performance-based clusters explain differences in league outcomes’ measured by average points accumulation, while controlling for potential relevant variables that would contribute to the validity of the statistical model to be utilised. The present section continues the work of the previous sections and builds on the conclusion reached about the existence of three performance groups and their significant

relationship with points accumulation. The use of multiple regression analysis here will quantify more formally the explanatory power of club cluster membership upon league outcomes.

Indeed, as opposed to Pearson's correlation test performed at Section 4.4 on the bivariate relationship between cluster membership and average points per league, it enables one to introduce control variables.

Two regression models have been used. The first one consisted of a simple regression where average points was only dependent on cluster membership; the second was a multiple regression where, in addition, average goals for, and average goals conceded have been included to control for attacking and defensive aspects. A check of regression assumptions of homoscedasticity, absence of multicollinearity and normality of errors will be presented before interpreting the results.

Figure 1: Q-Q plot of residuals - Simple Regression Model

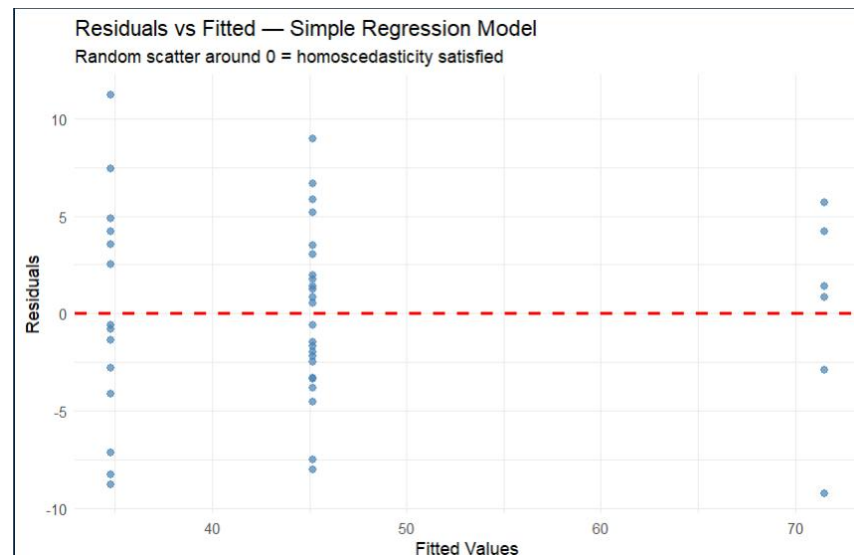


The data points mostly follow the red reference line within the range of 1 to 2 across the centre, so as far as they lie there the distribution appears to be normal - the majority of points fall there as such there does seem to be a generally normal distribution of the club residual data points around zero. Further inspection of the lower tail suggests there are a few points lying below the reference line (range -1 to 2.5 which in effect are -1 and +2.5) there seems to be a small set of clubs that produce more extreme negative results than would be expected if the results were normal. What this is the number of clubs who perform far worse than is expected from what cluster they are supposed to be in- most likely to be the clubs at the top/bottom of the cluster groups (this interpretation has yet to be confirmed and is to be checked with the cluster centres).

I noticed that there were good following of the red reference line towards the upper tail of the graph.

Although the lower tail deviates from normal this does not strongly affect my findings as it should be noted ANOVA's used with in this modelling framework is robust to slight deviations of normal distribution for large samples of this size; however the formal Shapiro-Wilk test on the residual plots will further determine the significance of the deviation.

Figure 2: Residual Vs fitted



The plot reveals three distinct vertical bands of residuals corresponding to the three fitted values of approximately 34.8, 45.2, and 71.5 — representing the predicted average points for the Struggling, Mid-Table, and Elite clusters respectively. This banding pattern is expected and inherent to regression models with categorical predictors, where every observation within the same cluster receives the same fitted value, producing vertical columns of residuals rather than a continuous scatter.

Examining the spread of residuals within each band, the Mid-Table cluster (fitted value ≈ 45.2) displays the widest vertical spread, ranging from approximately -7 to $+9$, which is consistent with the larger number of clubs in this group and the greater within-group variation observed earlier. The Struggling cluster (fitted value ≈ 34.8) shows a moderate spread, while the Elite cluster (fitted value ≈ 71.5) displays a relatively narrower spread, reflecting the internal consistency of the six elite clubs.

The unequal spread across the three bands suggests a mild violation of homoscedasticity the residual variance appears larger for the Mid-Table group than for the Elite group. However, this is a common and expected feature of regression models with cluster-based categorical predictors, where group sizes differ considerably. The formal Breusch-Pagan test result will confirm whether this visual pattern constitutes a statistically significant violation requiring corrective action.

Table 1: Durbin-Watson Test for Independence of Residuals — Simple Regression Model

Model	DW Statistic	p-value	Autocorrelation	Decision
Simple Regression (Avg_Pts ~ Cluster)	2.395	0.899	None	Assumption Met

Note: DW = Durbin-Watson statistic. Values close to 2 indicate no autocorrelation. A p-value greater than 0.05 confirms the absence of significant positive autocorrelation in the residuals, satisfying the independence assumption of the regression model.

The Durbin-Watson test was conducted to assess whether the residuals of the simple regression model are independent of one another — a fundamental assumption of ordinary least squares regression.

The test yielded a Durbin-Watson statistic of 2.395 and a p-value of 0.899, both of which confirm the absence of significant autocorrelation in the residuals. The DW statistic of 2.395 is close to the ideal value of 2, indicating negligible serial correlation, and the large p-value substantially exceeds the 0.05 threshold, providing no evidence to reject the null hypothesis of zero autocorrelation. The independence of residuals assumption is therefore satisfied, confirming that the average points values of individual clubs do not systematically influence one another in a way that would bias

the regression estimates. This is an expected finding given that the observations represent different clubs rather than repeated measurements over time, and supports the validity of the inferential conclusions drawn from the simple regression model.

Table 2: Model Fit Statistics — Simple Regression Model

Model	R ²	Adjusted R ²	F-statistic	p-value
Avg_Pts ~ Cluster_Name	0.846	0.839	F(2, 41) = 113.00	< 0.001 ***

*Note: R² = coefficient of determination; Adjusted R² accounts for the number of predictors in the model. Residual standard error = 5.006 on 41 degrees of freedom. Elite cluster serves as the reference category. *** p < 0.001.*

Table two presents the overall model fit statistics for the simple regression model in which average league points was regressed solely on performance group membership.

The model demonstrated exceptional explanatory power, with an R² of 0.846 indicating that cluster membership alone accounts for approximately 84.6% of the total variation in average league points across the 44 EPL clubs. The adjusted R² of 0.839, which penalises for the number of predictors included in the model, remained virtually identical to the unadjusted value — confirming that the high explanatory power is not an artefact of overfitting but reflects a genuinely strong relationship between performance group classification and points accumulation.

The overall model was statistically significant (F(2, 41) = 113.00, p < 0.001), confirming that the regression model explains a significantly greater proportion of variance in average points than would be expected by chance alone. These findings establish that performance-based cluster membership is an exceptionally powerful predictor of league outcomes, providing a strong empirical foundation for the

subsequent multiple regression analysis in which control variables are introduced to further examine the robustness of this relationship.

Table 3: R-Squared Comparison — Simple vs Multiple Regression Model

Model	Predictors	R ²	R ² Improvement
Model III (Simple Regression)	Avg_Pts ~ Cluster	0.8464	+18.58%
Model IV (Multiple Regression)	Avg_Pts ~ Cluster + Avg_GF + Avg_GA + Seasons_Played	0.9822	—

Note: R² = coefficient of determination. Model III includes cluster membership only. Model IV additionally incorporates average goals scored (Avg_GF), average goals conceded (Avg_GA), and seasons played as control variables. R² improvement reflects the gain in explanatory power from Model III to Model IV.

The simple regression model incorporating cluster membership as the sole predictor, achieved an R² of 0.846, confirming that performance group classification alone explains approximately 84.6% of the variation in average league points — an already substantial level of explanatory power. Model IV, the multiple regression model which extended this by incorporating average goals scored, average goals conceded, and seasons played as additional control variables, yielded a markedly higher R² of 0.982, indicating that the expanded model accounts for approximately 98.2% of the total variation in average points.

The addition of the three control variables produced an improvement of approximately 18.6 percentage points in R², demonstrating that while cluster membership is a powerful predictor of league outcomes in isolation, the inclusion of goal-related performance indicators and tenure length captures additional dimensions

of team performance that cluster assignment alone does not fully account for. The near-perfect explanatory power of Model IV suggests that average goals scored, average goals conceded, seasons played, and cluster membership together provide an exceptionally comprehensive account of the factors driving variation in league points accumulation across EPL clubs between 2000–01 and 2021–22. This improvement underscores the value of incorporating control variables to enhance the robustness and validity of the regression framework, as specified in Objective IV.

4.6: Discussion of findings:

This study set out to classify English Premier League clubs into distinct performance-based groups and to examine the relationship between these classifications and league points accumulation across the 2000–01 to 2021–22 seasons. The findings across all four objectives consistently point to a coherent and statistically robust pattern of performance stratification within the league.

With respect to the first objective, the K-Means clustering algorithm successfully identified three distinct performance tiers among the 44 clubs analysed — Elite, Mid-Table, and Struggling — based on average seasonal wins, draws, losses, goals scored, and goals conceded. The Elite cluster comprised six clubs, namely Manchester United, Chelsea, Arsenal, Liverpool, Manchester City, and Tottenham Hotspur, all of whom demonstrated sustained dominance across multiple seasons, consistent with the well-documented concentration of financial resources among a small group of top clubs in the Premier League era. The remaining clubs were distributed across the Mid-Table ($n = 24$) and Struggling ($n = 14$) clusters, with the latter characterised by brief EPL

tenures, high goals conceded, and consistently low points returns. The validity of this three-cluster solution was supported by strong Hopkins statistic values ranging from 0.70 to 0.83 and convergent evidence from both the elbow and silhouette methods.

Addressing the second objective, the descriptive statistics revealed substantial differences in average points accumulation across the three clusters — 71.5, 45.2, and 34.8 points per season for Elite, Mid-Table, and Struggling clubs respectively. These differences were confirmed as statistically significant by the one-way ANOVA ($F(2, 41) = 113.00, p < 0.001$), with an eta-squared of 0.85 indicating that cluster membership alone explained 85% of the total variation in league points. Post-hoc Tukey HSD comparisons confirmed that all three pairwise group differences were statistically significant, with the largest gap observed between Elite and Struggling clubs (mean difference = 36.7 points, $p < 0.001$), underscoring the vast performance chasm that separates the league's dominant sides from its most vulnerable.

Regarding the third objective, the Pearson correlation analysis yielded a strong negative coefficient of $r = -0.868$ ($p < 0.001, R^2 = 0.753$), confirming that as cluster ranking moves from Elite to Struggling, average points accumulation declines substantially and consistently. This strong association validates the construct validity of the clustering solution — the groups identified by the unsupervised algorithm align closely with actual league performance outcomes, despite points not having been used as a clustering input, reinforcing the argument that match outcome and goal-related indicators are highly effective proxies for overall league standing.

Finally, addressing the fourth objective, the regression analysis provided the most comprehensive assessment of the relationship between cluster membership and league outcomes. The simple regression model confirmed that cluster membership alone

explained 84.6% of the variation in average points ($R^2 = 0.846$, $F(2, 41) = 113.00$, $p < 0.001$), while the multiple regression model incorporating average goals scored, average goals conceded, and seasons played as control variables elevated this to 98.2% ($R^2 = 0.982$) — an improvement of approximately 18.6 percentage points. Taken together, these findings demonstrate that performance-based classifications derived from unsupervised machine learning provide a robust, parsimonious, and statistically powerful framework for understanding and predicting league outcomes in the English Premier League.

